DO NOW — RECALL

Lesson 102 · Choosing RISC, CISC, GPU and Multicore Sy

Name: _____ Class: _____ Date: _____

1. What does RISC primarily describe?

2. What does a GPU excel at?

3. Can a RISC CPU be multicore?

4. Graph edges A–B, A–C, B–D: BFS from A with B before C?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A reduced/regular instruction-set design approach.

2. Model answer

Large quantities of similar parallel operations.

3. Model answer

Yes. ISA and core count are independent characteristics.

4. Model answer

A,B,C,D.

By the end, you can…

• RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

• A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

• Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

Vocabulary — know these terms

ISA — Instruction Set Architecture.

accelerator — Hardware specialised for particular workloads.

latency — Delay of a task.

throughput — Amount of work per unit time.

NEW LEARNING

Read, annotate and explain

Understand

RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

Apply

A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

Evaluate

Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Recommend hardware for battery-powered sensor making a small control decision each second.

Explain the choice

A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain why RISC/CISC and multicore are not competing mutually exclusive categories. [6]

Question 2

Select suitable processor features for (a) small battery sensor (b) large parallel graphics processing. Justify. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate CPU vs GPU investment for a school's interactive office tasks and high-volume video editing. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every multicore processor must use a CISC instruction set.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.